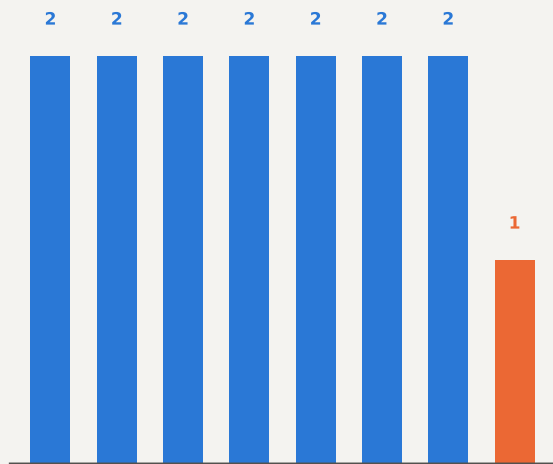


Inverse Banzhaf voting-design

given a target power distribution, construct a realizing rule — or certify that none exists



target ψ^n : $n-1$ players 2x, one player 1x

Two rule classes, opposite verdicts

Simple games (W-family)



explicit realizing game for every $n \geq 6$

Weighted majority games

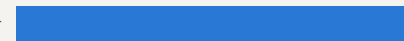


CP-SAT certifies: none realizes the target, $6 \leq n \leq 10$

Exact case study

weighted EU Council (EEC-6/9/10, EU-15)

actual power



$L_1 \approx 0.34-0.42$

certified redesign



$L_1 \approx 0.11$ (3x closer)

real councils are far from egalitarian

no weighted redesign reaches the target

certified optimal redesign improves ~3x